

States: Which squares are dirty/clean
Position of vacuum

Action: Left, Right, Clean

Reward: $r(s, a) = \# \text{ of clean squares in } s$

$$Q(s, a) = \theta_1 f_1(s, a) + \theta_2 f_2(s, a)$$

$f_1(s, a) = \# \text{ of clean squares}$

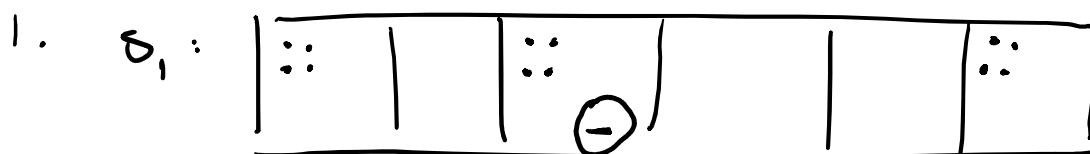
$f_2(s, a) = \text{vacuum in dirty square} \Leftrightarrow a \text{ is Clean}$

"Deep Q-Learning"

$$\gamma = 0.9$$

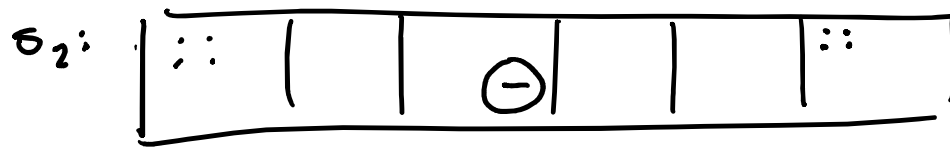
$$\alpha = 0.1$$

0. $\theta_1 = 1, \theta_2 = 1$



Take action $a_1 = \text{Clean}$.

We end up



$$\begin{aligned} 2. \quad Q(s_1, a_1) &= 1 \cdot 3 + 1 \cdot 1 \\ &= 4 \end{aligned}$$

$$\begin{aligned} Q(s_2, L) &= 1 \cdot 4 + 1 \cdot 1 \\ &= 5 \end{aligned}$$

$$\begin{aligned} Q(s_2, R) &= 1 \cdot 4 + 1 \cdot 1 \\ &= 5 \end{aligned}$$

$$\begin{aligned} Q(s_2, C) &= 1 \cdot 4 + 1 \cdot 0 \\ &= 4 \end{aligned}$$

$$\begin{aligned} 3. \quad L &= \frac{1}{2} (Q(s_1, a_1) - \gamma)^2 \\ \gamma &= r(s_1, a_1) + \gamma_{\max_{a'}} Q(s_2, a') \end{aligned}$$

$$\begin{aligned} \gamma &= 3 + 0.9 \cdot 5 \\ &= 7.5 \end{aligned}$$

$$L = \frac{1}{2} (4 - 7.5)^2$$

$$= 6.125$$

$$\begin{aligned}\frac{\partial L}{\partial \theta_i} &= (Q(s_i, a_i) - y) \cdot \frac{\partial}{\partial \theta_i} Q(s_i, a_i) \\ &= (Q(s_i, a_i) - y) \cdot f_i(s_i, a_i)\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial \theta_1} &= (4 - 7.5) \cdot 3 \\ &= -10.5\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial \theta_2} &= (4 - 7.5) \cdot 1 \\ &= -3.5\end{aligned}$$

$$4. \quad \theta_i \leftarrow \theta_i - \alpha \frac{\partial L}{\partial \theta_i}$$

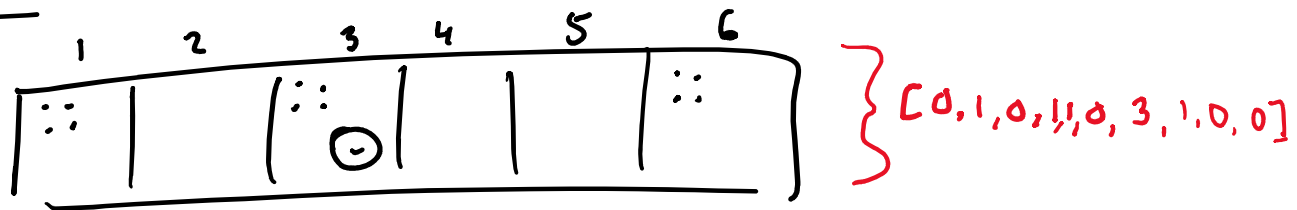
$$\begin{aligned}\theta_1 &\leftarrow 1 - 0.1 \cdot (-10.5) \\ &\leftarrow 2.05\end{aligned}$$

$$\begin{aligned}\theta_2 &\leftarrow 1 - 0.1 \cdot (-3.5) \\ &\leftarrow 1.35\end{aligned}$$

5. Repeat.

Example

Example



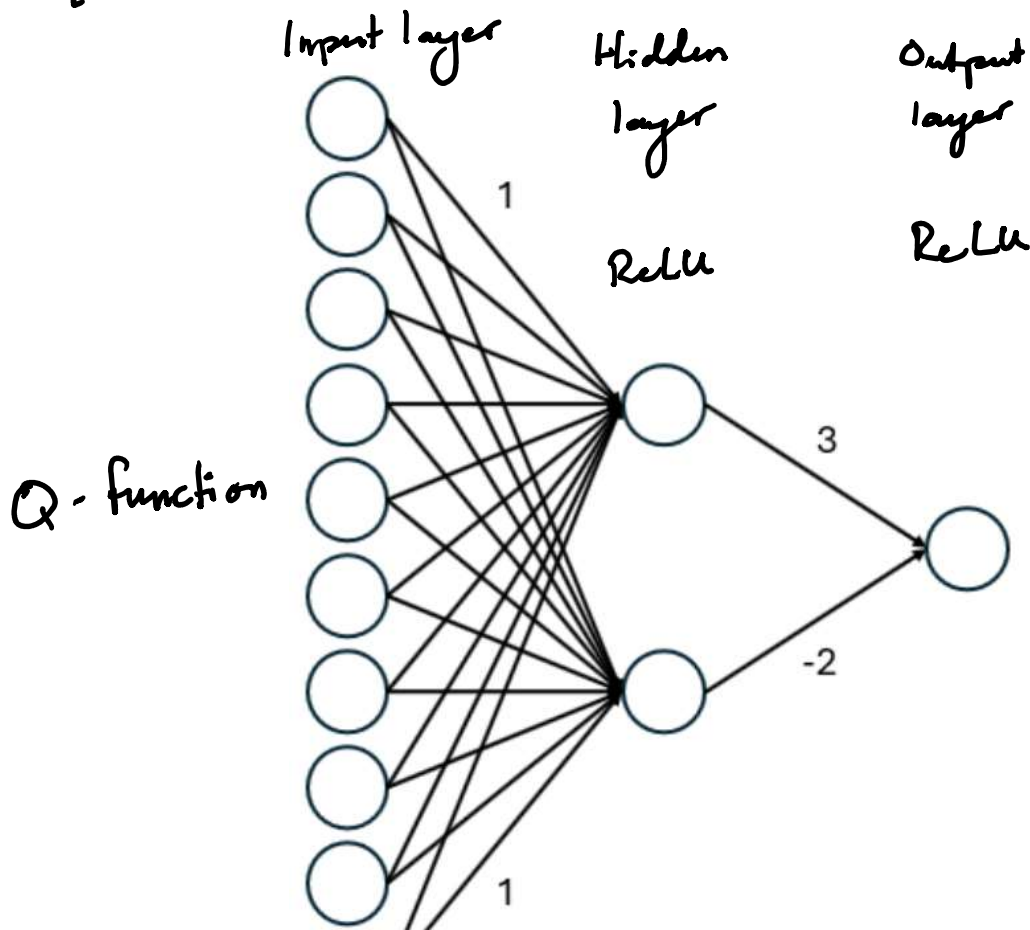
Let's represent state-action pairs

$$[c_1, c_2, c_3, c_4, c_5, c_6, p, a_L, a_R]$$

c_i - indicates whether i^{th} square is clean

p - position of vacuum

a_i - indicates whether i^{th} action was taken





Deep Q-Learning

$$\gamma = 0.9$$

$$\alpha = 0.1$$

0. Weights for bias terms 0.

Weights in hidden layer 1.

Weights for output layer $[3, -2]$.

$$1. \quad S_1: \begin{bmatrix} :: & & :: & \ominus & & & :: \end{bmatrix}$$

Action: $a_1 = \text{Clean}$

$$S_2: \begin{bmatrix} :: & & & \ominus & & & :: \end{bmatrix}$$

$$2. \quad Q(S_1, a_1) = [0, 1, 0, 1, 1, 0, 3, 1, 0, 0] \\ = 7$$

$$Q(S_2, L) = [0, 1, 1, 1, 1, 0, 3, 0, 1, 0] \\ = 8$$

$$Q(S_2, R) = [0, 1, 1, 1, 1, 0, 3, 0, 0, 1] \\ = 8$$

$$Q(S_1, C) = [0, 1, 1, 1, 1, 0, 3, 1, 0, 0]$$

$$Q(s_2, c) = [0, 1, 1, 1, 1, 0, 3, 1, 0, 0] \\ = 8$$

$$3. \quad L = \frac{1}{2} (Q(s_1, a_1) - Y)^2 \\ Y = r(s_1, a_1) + \gamma \max_{a'} Q(s_2, a')$$

$$Y = 3 + 0.9 \cdot 8 \\ = 10.2$$

$$L = \frac{1}{2} (7 - 10.2)^2 = 5.12$$

$$\frac{\partial L}{\partial \theta_i} = (Q(s_1, a_1) - Y) \cdot \frac{\partial}{\partial \theta_i} Q(s_1, a_1)$$

For the output layer:

$$\frac{\partial Q}{\partial \theta_i} = \text{ReLU}'(\sum w_i y_i) \cdot y_i$$

y_i is the inputs to the output layer

w_i is the weights for the output layer

$$Q = \text{ReLU}(\sum w_i y_i)$$

$$\frac{\partial Q}{\partial w_i} = \text{ReLU}'(\sum w_i y_i) \cdot y_i \\ = 7$$

$$\frac{\partial L}{\partial \theta_i} = (7 - 10.2) \cdot \underline{7} = -22.4$$

$$\frac{\partial L}{\partial w_1} = (7 - 10.2) \cdot \underline{7} = -22.4$$

$$\frac{\partial L}{\partial w_2} = (7 - 10.2) \cdot \underline{7} = -22.4$$

$$\frac{\partial L}{\partial v_i} = \dots ? \quad \text{Use backprop.}$$

4. Update by gradient descent.

$$\theta_i \leftarrow \theta_i - \alpha \frac{\partial L}{\partial \theta_i}$$

$$\begin{aligned} w_1 &\leftarrow 3 - 0.1 \cdot (-22.4) \\ &\leftarrow 5.24 \end{aligned}$$

$$\begin{aligned} w_2 &\leftarrow -2 - 0.1 \cdot (-22.4) \\ &\leftarrow 0.24 \end{aligned}$$

$$\theta_i \leftarrow \dots ?$$

5. Repeat.

Policy Gradient Methods.

Idea: Approximate π as a neural network.

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Learn parameters of π to maximize rewards.

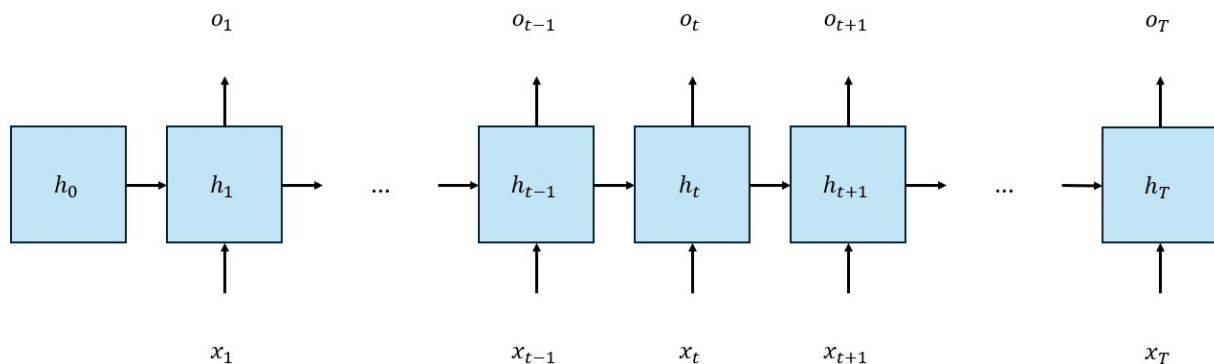
Attention

Sequence-to-sequence tasks:

Input is a sequence
Output is a sequence.

Examples

Language translation
Video labelling



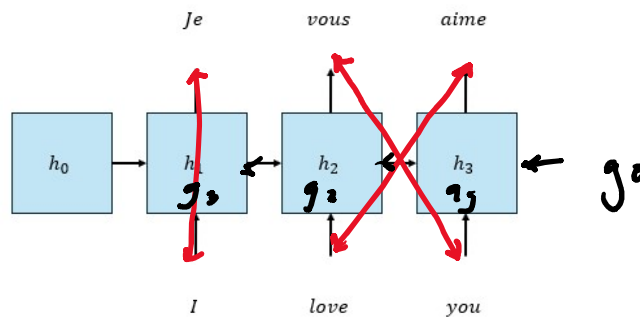
Issues:

1. Cannot well model long-term dependencies.

2. Cannot parallelize.

3. Input + output must have same length.
Input + output must have same ordering.

3. Input + output must have same ordering.
4. Assumption input + output have same ordering.



RNN Encoder + Decoder

